

3.13.1 Discrete semiconductor devices (A-level only)

3.13.1.1 MOSFET (metal-oxide semiconducting field-effect transistor) (A-level only)

Content

Simplified structure, behaviour and characteristics.

Drain, source and gate.

V_{DS} , V_{GS} , I_{DSS} , and V_{th}

Use as a switch, use as a device with a very high input resistance.

Use in N-channel, enhancement mode only is required.

3.13.1.2 Zener diode (A-level only)

Content

Characteristic curve showing zener breakdown voltage and typical minimum operating current.

Anode and cathode.

Use with a resistor as a constant voltage source.

Use to provide a reference voltage.

Use as a stabiliser is not required.

Characteristic curves and spectral response curves.

Use in photo-conductive mode as a detector in optical systems.

Use with scintillator to detect atomic particles.

3.13.1.4 Hall effect sensor (A-level only)

Content

Use as magnetic field sensor to monitor attitude.

Use in tachometer.

Principles of operation are not required.